

9. Cobalt forms a range of complex ions. Two of these are  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  and  $[\text{CoCl}_4]^{2-}$ .

(a) Draw the structures of the  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  and  $[\text{CoCl}_4]^{2-}$  ions. [2]

(b) Explain why the  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  ion is coloured. [3]

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- (c) A student was given a pink-coloured solution containing  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  ions. Upon addition of hydrochloric acid the solution turned blue as  $[\text{CoCl}_4]^{2-}$  ions formed according to the equilibrium shown below.



Aqueous silver nitrate was added to the solution containing  $[\text{CoCl}_4]^{2-}$ .

State and explain the observation(s) expected.

[4]

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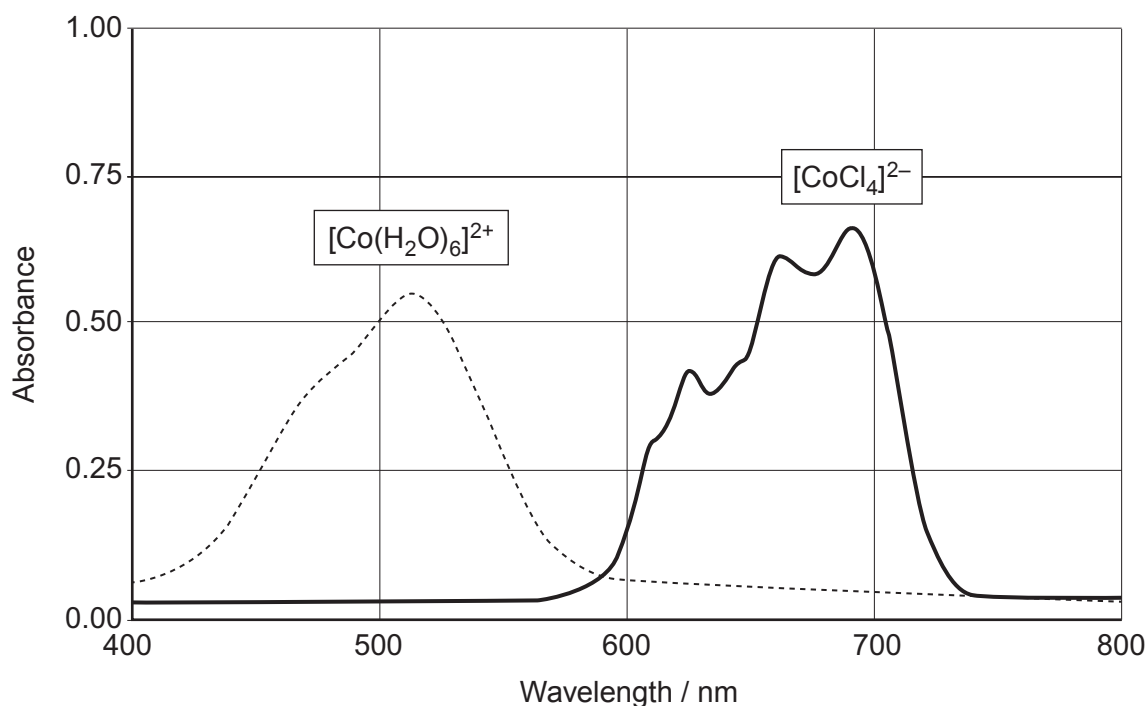
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- (d) These two cobalt-containing ions absorb different frequencies of light. The spectra below show the absorbance at different wavelengths of visible light by the two ions.



- (i) The maximum absorbance for the  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  ion occurs at a wavelength of 518 nm. Calculate the energy of this absorbance in  $\text{kJ mol}^{-1}$ . [3]

Energy = .....  $\text{kJ mol}^{-1}$

- (ii) A colorimeter was used to study the equilibrium between the two ions.

Suggest, giving a reason, an appropriate wavelength to use for the experiment. [1]

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- (iii) Write the expression for the equilibrium constant,  $K_c$ , for this reaction **giving its unit**. [2]



$K_c =$

Unit .....

- (iv) A student carries out an experiment starting with separate non-aqueous solutions of  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  and  $\text{Cl}^-$ .

When the two solutions are mixed the initial concentrations of both  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  and  $\text{Cl}^-$  in the mixture are  $0.720 \text{ mol dm}^{-3}$ . When the mixture reaches equilibrium the concentration of the water is  $0.744 \text{ mol dm}^{-3}$ .

Calculate the value of the equilibrium constant,  $K_c$ . [3]

$K_c =$  .....

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